CS611 Loan Default End to End Pipeline

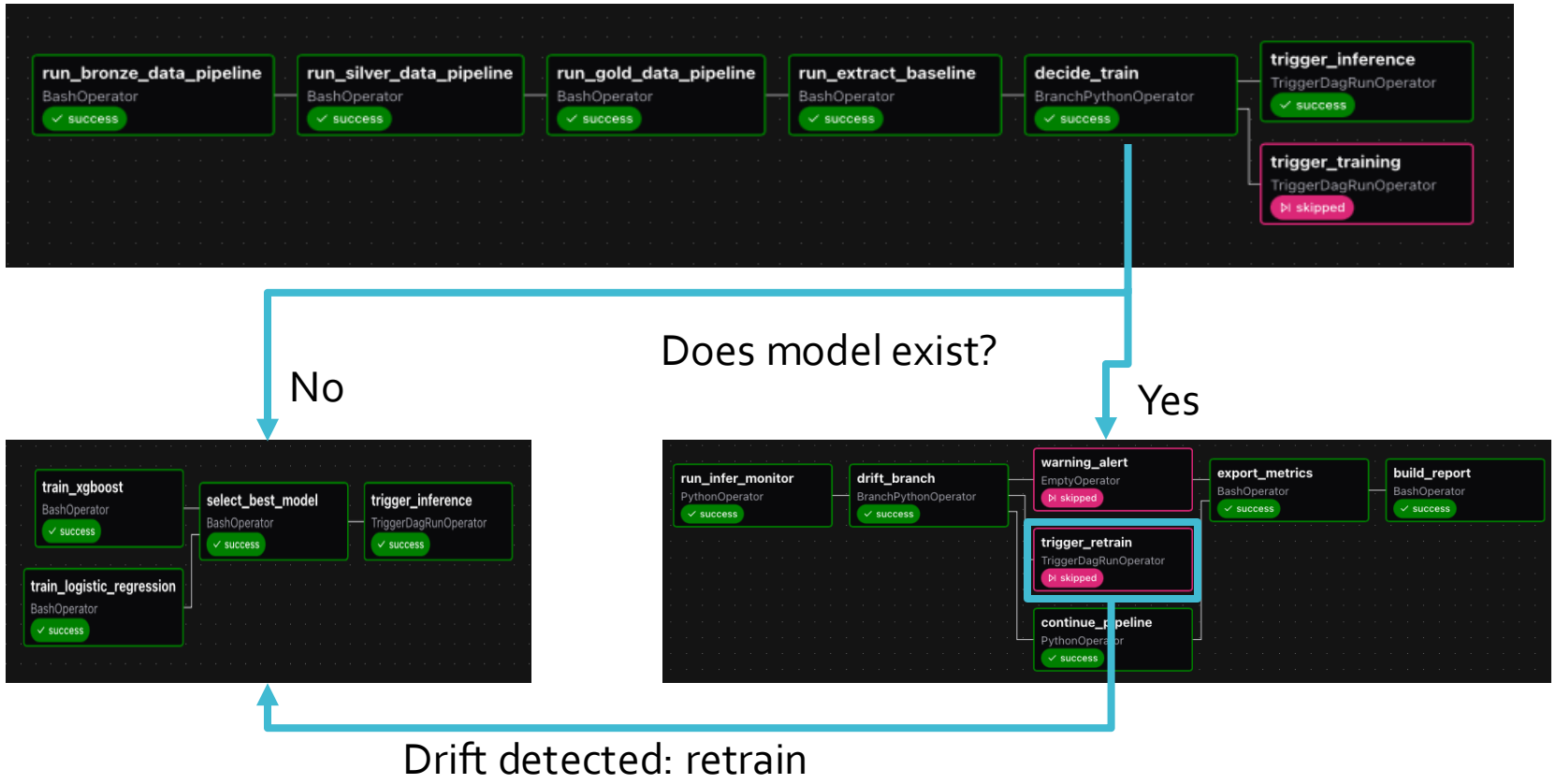
Wyman Lee

Background & Objective

- **Key challenge**
  - Mitigate loan default risk by accurately assessing applicant creditworthiness at the point of application
- **Objective**
  - Develop machine learning model to determine if customer will default on a loan before it is approved
- **Setup**
  - ETL execution via Airflow UI
  - Host: mle_asm2-airflow-apiserver-1
  - Port: 8080
  - DAG: etl_pipeline

End to end ML pipeline

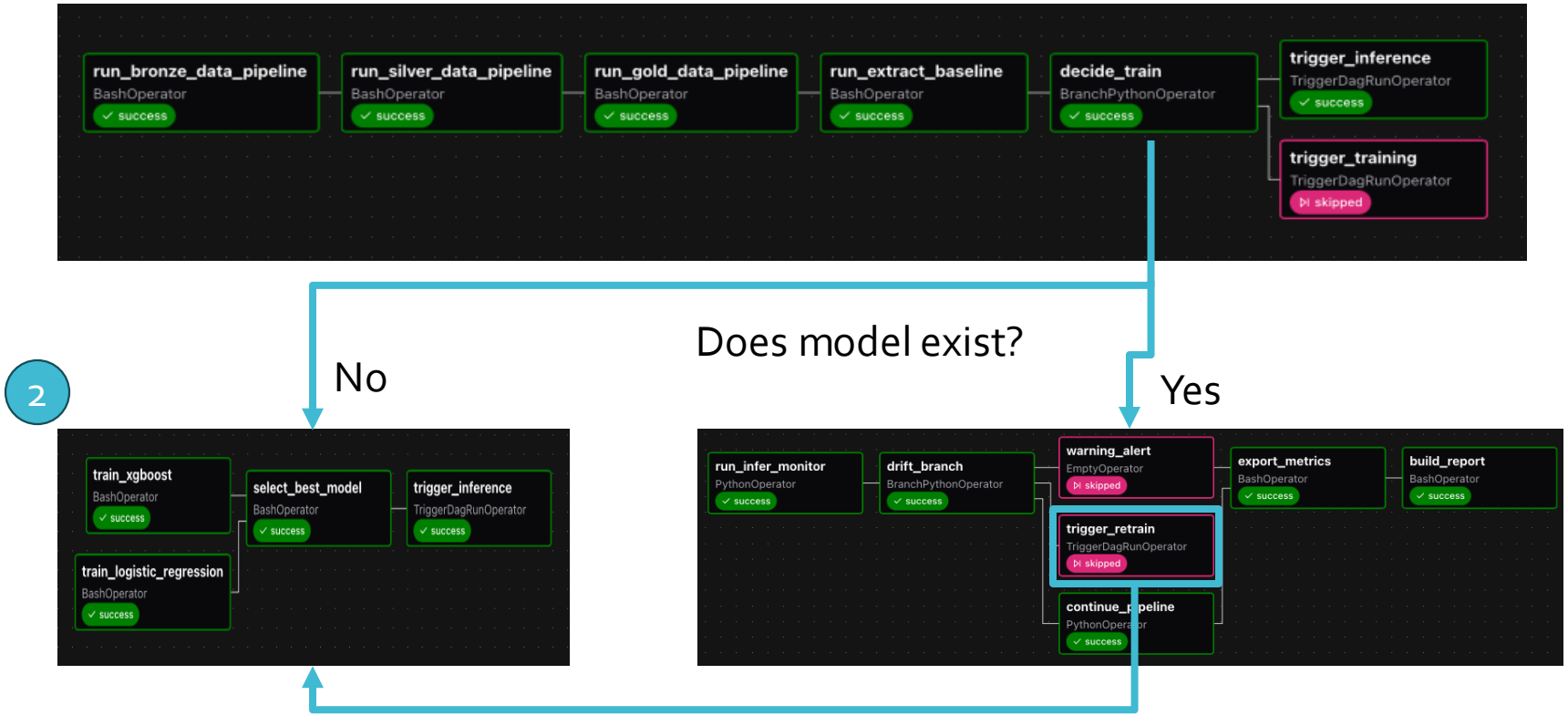
1



1

Task (Node)	Purpose
run_extract_baseline	Snapshot latest features as "production baseline" for drift comparisons
decide_train	Check for existing model artifacts; branch to train vs. infer
trigger_training	Kick off full model-training DAG when no production model exists
trigger inference	Kick off inference+monitoring DAG when a baseline model is already in place

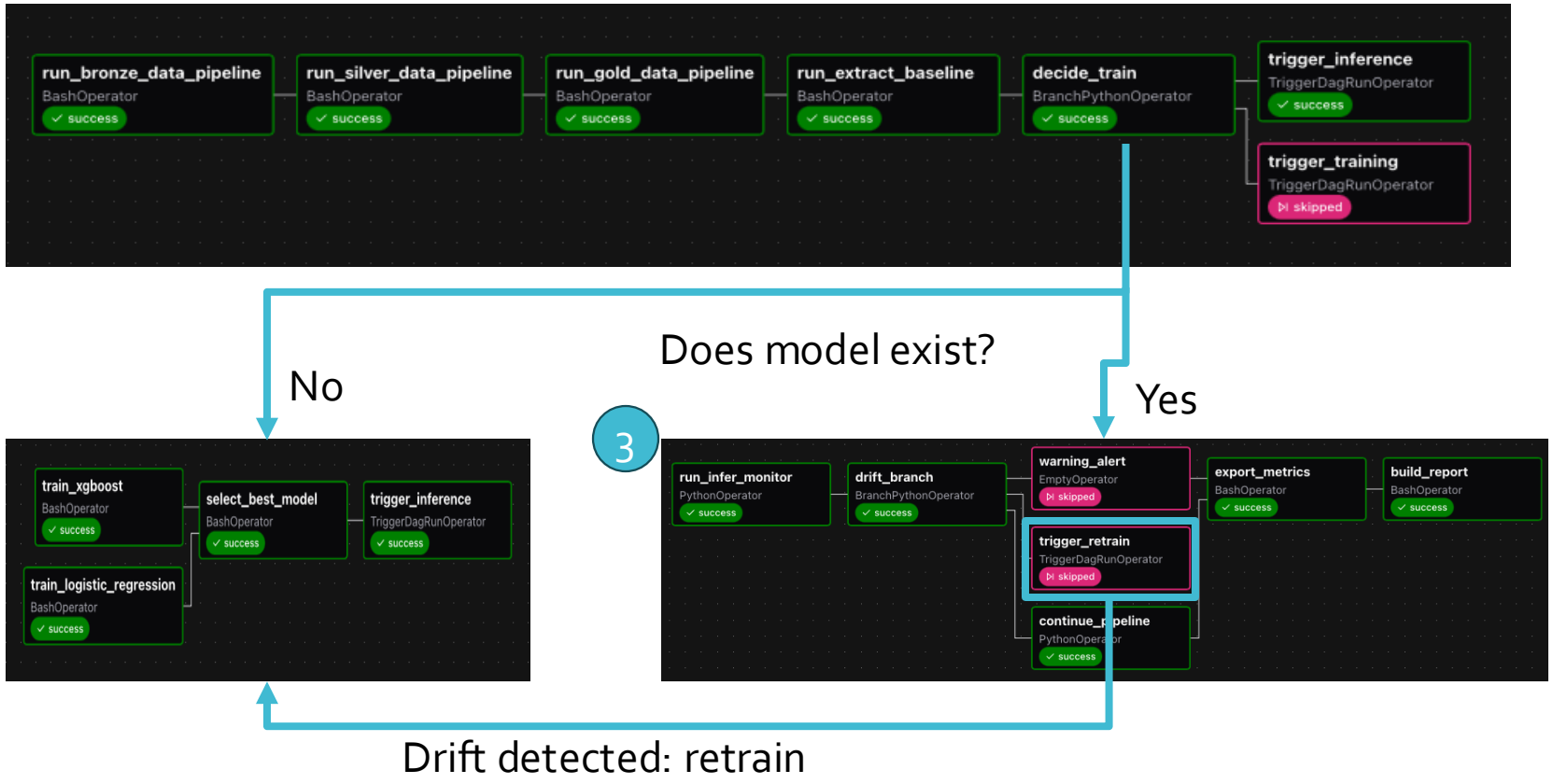
End to end ML pipeline



2

Task (Node)	Purpose
train_logistic_regression	Fit baseline LR model, log metrics (ROC-AUC)
train_xgboost	Fit boosted-tree model, log metrics (ROC-AUC)
select_model	Compare candidate models on validation/holdout metrics and choose the best performer.
trigger inference	monitoring DAG when a baseline model is already in place

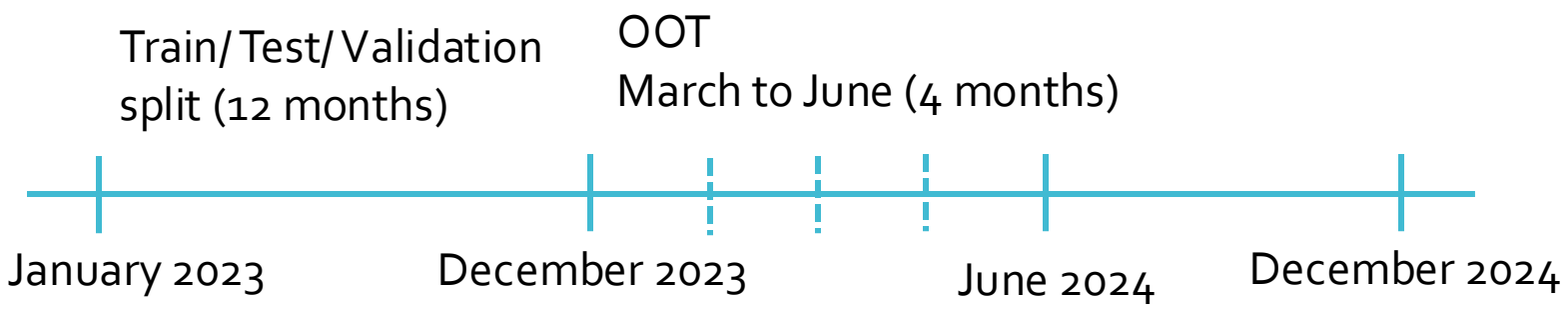
End to end ML pipeline



3

Task (Node)	Purpose
run_infer_monitor	Score new snapshot with prod model; compute PSI per feature; log metrics
drift_branch	Evaluate PSI_max vs. threshold → route to safe/warning/critical paths
continue_pipeline	(Safe) No action needed; proceed to metrics export
warning_alert	(Warning) Trigger warning logs

Model Training & Inference Strategy



Data Splitting Strategy

- 18-month window: Jan 2023–Jun 2024
- Train (Jan 2023- Dec 2023, 12 months to capture seasonality)
- Validation (January 2024, to capture post holiday transaction surge when tuning hyperparameters)
- Test (February 2024, last month before deployment to capture unbiased performance estimate)

Data Horizon

- All labeled snapshots **Jan 1, 2023 – pipeline trigger date**
- Ensures full exposure to seasonal spending & fraud campaigns

Model Selection Strategy

Models chosen (Trained in parallel):

Model	Why it was chosen
Logistic regression	Simple, interpretable, efficient baseline for binary classification with linearly separable data
XGBoost	High performance, captures complex patterns, robust to nonlinear relationships

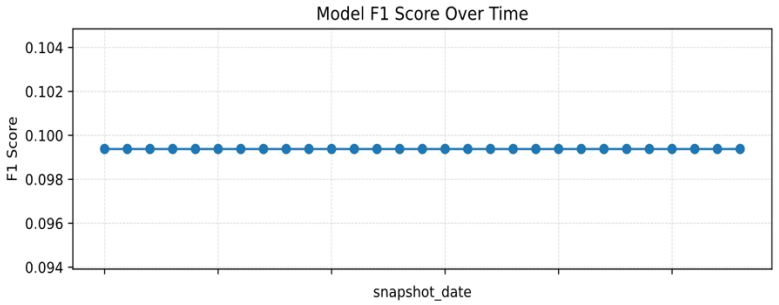
Evaluation metrics:

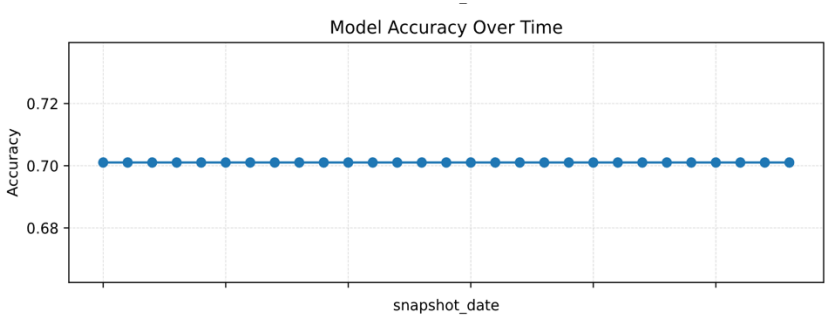
Metric	Why it was chosen
ROC-AUC	Robust to class imbalance, provides a comprehensive view of model performance across all classification thresholds, widely used industry standard

Data and Labels:

	Why it was chosen	Source	Columns
Features	Captures detailed customer behavior and interactions, crucial for predictive modeling	data/feature clickstream	prefixed with "fe_"
Labels	Customer_ID • Uniquely identifies customers across datasets. • Ensures correct alignment of features with labels, maintaining dataset integrity Snapshot_Date • Aligns features and labels temporally • Prevents data leakage, enhancing model reliability and real-world performance	datamart/gold/label_store	Customer_ID, snapshot_date, label

Model Monitoring Strategy





- **Key Metrics Tracked**
  - F1 (Binary): prioritizes minority-class recall/precision balance
  - Accuracy: sanity-check for broad correctness
  - PSI (Population Stability Index): flags distributional drift at the feature level
  - Track model behavior for both backfill[1] and live inference
- **Monitoring strategy[2]**:
  - Inference period: March to June 2024 (OOT period)
  - Feature snapshot: Gold datamart snapshots taken at start of each month
  - Best model: Selected monthly based on OOT predictions from unseen test data

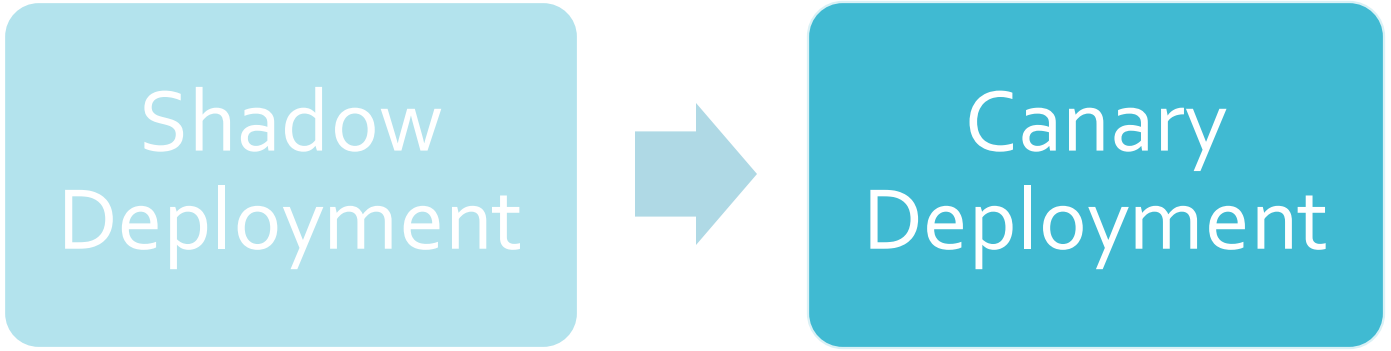
Deployment Strategy

Shadow Deployment → Canary Deployment

Task: run_infer_monitor
What it does: Runs the production model on **feature data from a specified snapshot date,** which is passed via the DAG configuration

Metric	Test for	Why
psi_max	Detects data drift: • High PSI (≥ **0.20**) triggers retrain • PSI between **0.10 to <0.20** triggers warning • PSI < **0.10** is considered safe	High PSI means **current production data significantly deviates from the baseline/training distribution,** risking model degradation
Prediction mean	Bugs in preprocessing, input features, or model loading	Checks the **average predicted probability** for the current snapshot. **close to 1 or close to 0 suggest Input feature corruption** (e.g., incorrect scaling, bad feature joins)
Prediction standard deviation	schema mismatch, nulls, or input errors	Measures the **spread of predictions**. Close to 0: Indicates near-uniform predictions (all 0s or all 1s), suggesting potential issues. Healthy models should show variability matching the training data behavior.

Deployment Strategy



Task: drift_branch
Trigger condition: PSI ratio between 0.5 and 1.0
What it does:

- Sends a **warning alert** about emerging data drift.

Data Scope: Pipeline continues running on the same snapshot date.

- Operates on feature data from the snapshot date provided at DAG trigger
- Example: If snapshot = 2024-06-01, all drift checks, metrics exports, and reports apply to **June 2024 data**

Outputs:

- **export_metrics:** Saves model metrics for the snapshot date.
- **build_report**: Generates a performance report tied to the same date.

Model Governance

Framework

- **Centralized Artifacts**

Unified storage of binaries, schemas & configs with metadata for consistency.

- **Automated Selection**

Evaluate on out-of-time data (F1/accuracy) and promote the top model.

- **Version & Registry**

MLflow tracking for reproducible, auditable deployments and rollbacks.

Refresh SOP

- Trigger conditions
  - when PSI ≥ 0.20
  - When new product launches & regulatory changes
- Retrain guidelines:
  - Collect latest 12 months training data
  - Deploy only if new model outperforms in OOT evaluation

Deployment SOP

- Monthly scoring via DAG
- Predictions stored as .parquet in datamart (Gold)

Conclusion

- End-to-end **explainable** pipeline
  - Model training, inference, monitoring and layered deployment
  - Integration with MLFlow for reproducibility
  - Use of industry recognized metrics for clean audits
- Model performance & monitoring
  - Performance metrics: Accuracy, F1 (binary)
  - Data drift monitoring with PSI
  - Prediction mean and standard deviation for anomaly detection
- Scalable and robust
  - Modular airflow DAGs for easy switching and scaling up of data and modelling workflows
- Future work
  - Real-time Inference with FastAPI and Redis-backed Online Feature Store